

Graph-RL-based Rendezvous Planning for Multiple Patrolling USVs under Current and Heading Constraints

Motivation & Problem

- Efficient rendezvous planning is required to inspect multiple patrolling USVs, particularly when different priorities are assigned to individual USVs.
- Ocean currents and heading constraints make travel time dependent on both position and direction, distinguishing this problem from conventional rendezvous planning.
- This study proposes an integrated framework that combines constrained path planning with learning-based rendezvous optimization.

Problem Formulation

Objective: $\min_{\Pi} [T_N + \sum_{\tau=1}^N \max(0, T_{\tau} - d_{\pi_{\tau}})]$

- Π : rendezvous order of patrolling USVs, d_i : priority-based due date for USV i
- T_{τ} : earliest feasible rendezvous time at decision step τ , jointly determined by the current state, patrol trajectory, and constrained travel time
- Each scheduling decision requires repeated path planning under ocean currents and heading constraints for multiple moving rendezvous candidates, posing a significant computational challenge.

Key Contributions

- A graph-based path-planning method using multi-stage maneuver approximation to account for ocean currents and heading constraints
- A graph neural network architecture that encodes spatiotemporal relationships among rendezvous candidates for priority-aware planning

PROPOSED APPROACH

Current-Aware Navigation

- Grid graph-based navigation for large maritime areas
- Current-aware edge costs with heading-constrained connectivity
- Multi-stage maneuver approximation for large heading changes, θ_{max}
- Hybrid path-tracking experiments to estimate θ_{max}

Rendezvous Planning

- Represent discretized rendezvous points as graph nodes linked by earliest feasible transitions
- Gated GCN encoding with travel time and due date as edge features
- PPO-based policy learning for sequential rendezvous decisions
- Direct travel-time prediction to reduce repeated path-search computation

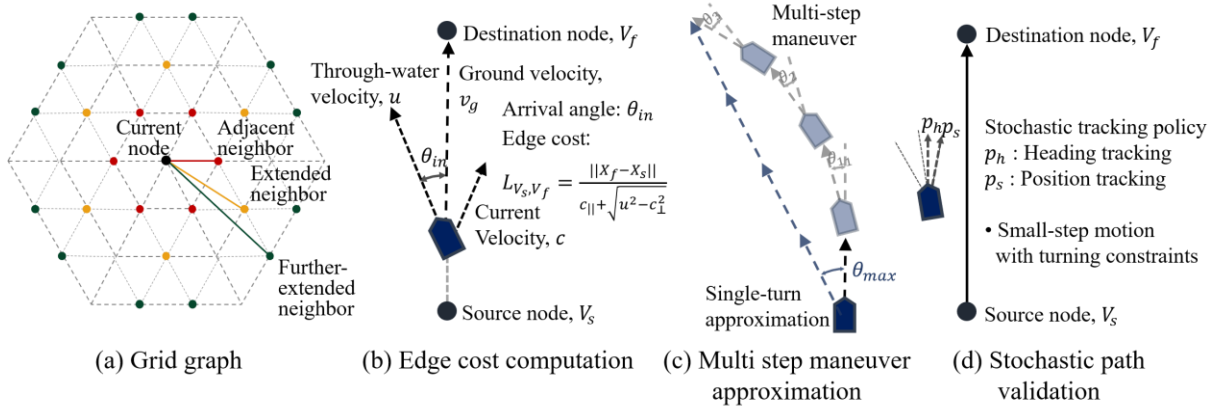


Figure 1. Current-Aware Path Planning

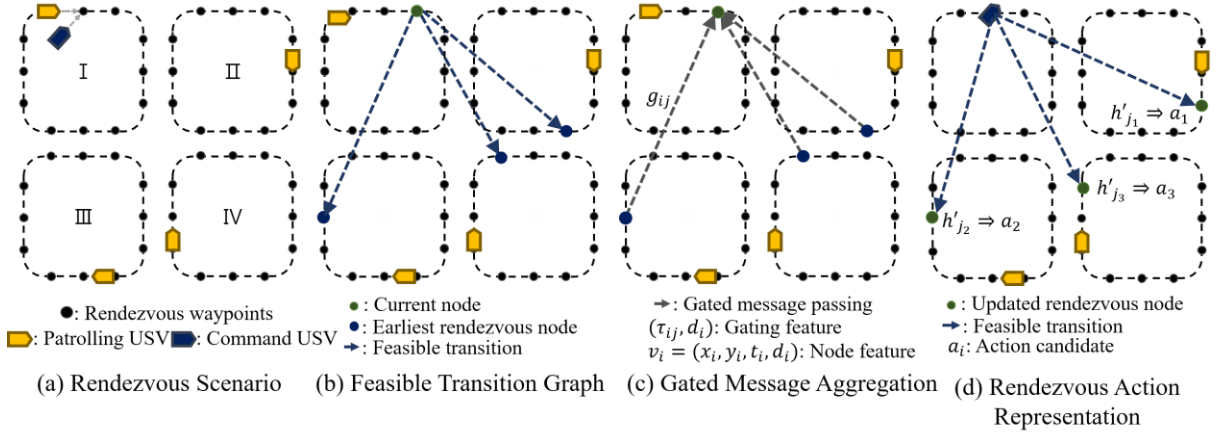


Figure 2 Rendezvous Planning

Key Result

Table 1. Comparison of Travel-Time Computation Methods

Method	Surrogate Prediction	Euclidean A*	Surrogate-guided A*	Dijkstra
Computation time	0.001s	0.195s	1.652s	1.876s
Gap from Dijkstra	3.023%	2.113%	0.883%	0.0%
Path Output	No	Yes	Yes	Yes

Table 2. Average Scaled Objective Value of each Algorithm

Algorithm	12-2T	<u>12-3T</u>	12-4T	16-2T	16-3T	16-4T	24-2T	24-3T	24-4T
Graph-RL	3.72 ± 0.7	4.16 ± 0.8	4.44 ± 1.1	4.39 ± 0.6	5.56 ± 1.2	6.02 ± 1.4	7.1 ± 2.2	7.62 ± 1.5	8.92 ± 1.9
MLP-RL	3.80 ± 0.7	4.42 ± 0.9	4.62 ± 1.0	4.66 ± 0.6	5.79 ± 1.0	6.24 ± 1.3	7.81 ± 2.3	8.31 ± 1.3	9.43 ± 1.5
Greedy	4.10 ± 1.0	4.79 ± 1.3	5.32 ± 1.5	5.15 ± 1.2	6.53 ± 1.6	7.02 ± 1.8	9.53 ± 3.7	10.2 ± 2.6	11.6 ± 2.6
PCNN	3.86 ± 0.6	4.78 ± 1.5	6.40 ± 2.3	4.67 ± 0.9	6.52 ± 1.8	9.05 ± 3.6	8.67 ± 4.3	9.92 ± 3.2	15.5 ± 6.8
Rolling Greedy	3.83 ± 0.7	4.27 ± 1.0	4.85 ± 1.2	4.79 ± 0.9	5.93 ± 1.5	6.43 ± 1.5	8.15 ± 2.0	8.40 ± 1.7	9.43 ± 1.7

Table 3. Relative Improvement over Baseline Method

Baseline Comparison	12-2T	<u>12-3T</u>	12-4T	16-2T	16-3T	16-4T	24-2T	24-3T	24-4T	Total
Graph-MLP	1.21%	4.53%	2.84%	5.15%	3.38%	2.55%	8.39%	7.14%	4.87%	4.45%
p-value	0.340	0.0079	0.124	0.0008	0.0818	0.149	0.0001	0.0018	0.0207	< 0.001

* N-MT: problem setting with N patrolling USVs and M Tier priority distribution

* PCNN: Priority-Constrained Nearest-Neighbor rule, * Rolling greedy: 2 steps rolling greedy

* Dark/light gray: performance advantage at the 5%/10% significance level, based on pairwise t-test

Grid-graph search based path planning

- Dijkstra provides the exact shortest-path solution on the formulated grid graph, but its repeated use is computationally expensive. We therefore compare several approximation methods, as summarized in Table 1.
- A decision-tree-based surrogate predictor offers a large reduction in computation time with limited accuracy loss, but does not provide an explicit path. Euclidean A* also reduces computation time while maintaining acceptable accuracy and returning an explicit path.
- The surrogate predictor is used during RL training for efficient travel-time estimation, whereas Euclidean A* is used for final evaluation.

Graph-RL based rendezvous planning

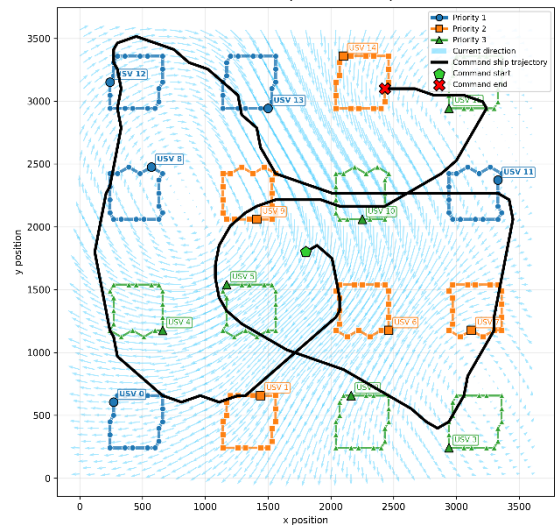
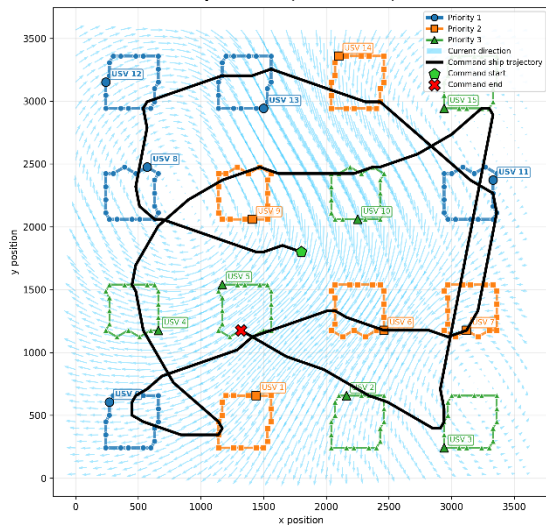
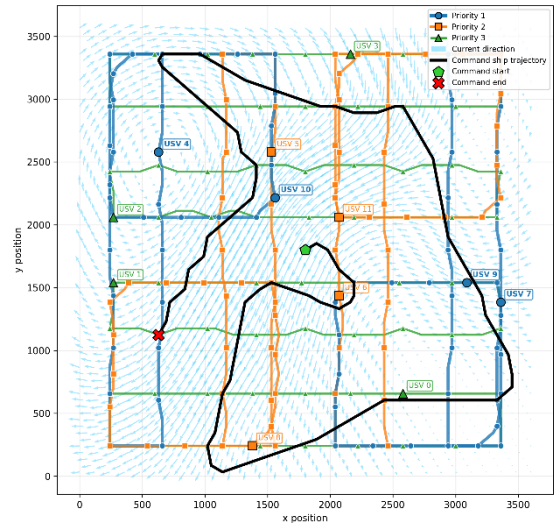
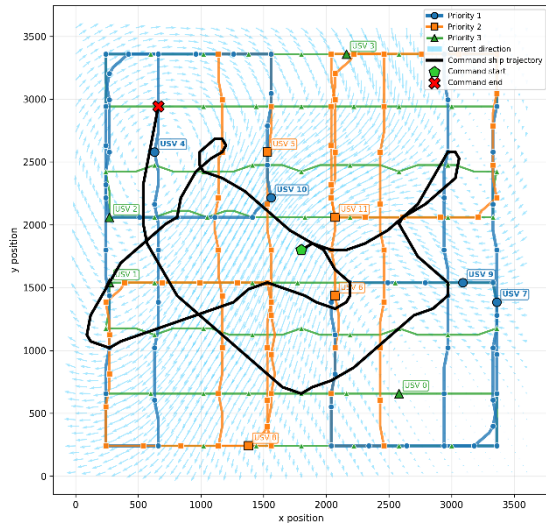
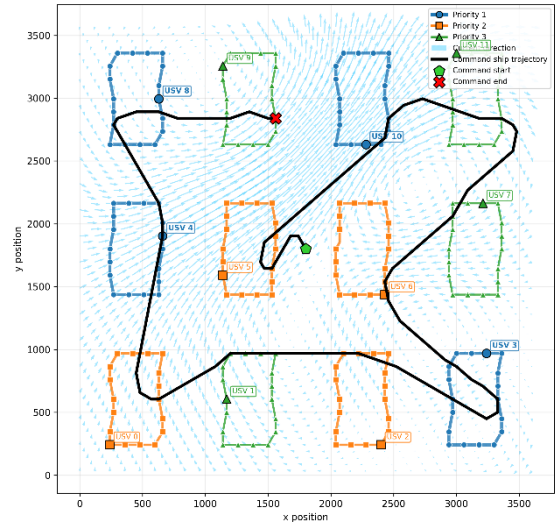
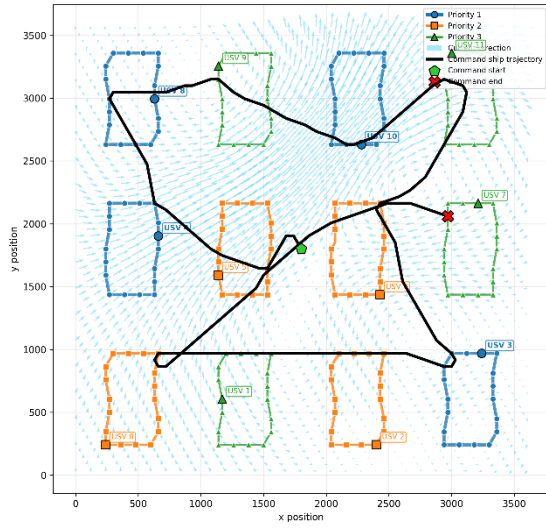
- Graph-RL and MLP-RL were trained only on the 12-USV, 3-tier (12-3T) distribution and evaluated across different fleet sizes and priority distributions, with randomized priorities, initial positions, and patrol directions
- MLP-RL serves as a strong learning-based heuristic using only local USV information.
- Graph-RL outperformed MLP-RL across all 3-tier settings and showed favorable trends on the extended 2- and 4-tier distributions, with larger gains as the fleet size increased.
- These results suggest that explicitly modeling relationships among spatiotemporal rendezvous candidates becomes increasingly effective as the number of feasible alternatives grows.

Discussion

- Figure 3 presents representative route solutions generated by Graph-RL and MLP-RL.

* Priority 1 denotes the highest priority, whereas Priority 3 denotes the lowest.

- Both methods produce high-quality routes that prioritize high-priority USVs while opportunistically rendezvousing with multiple lower-priority USVs along the way.
- The route examples suggest that Graph-RL tends to favor earlier service of high-priority USVs while opportunistically incorporating lower-priority rendezvous.



Graph RL (16-3T-a)

MLP-RL (16-3T-a)

Figure 3. Rendezvous Route Comparison: Graph-RL vs. MLP-RL

Appendix: Training curves

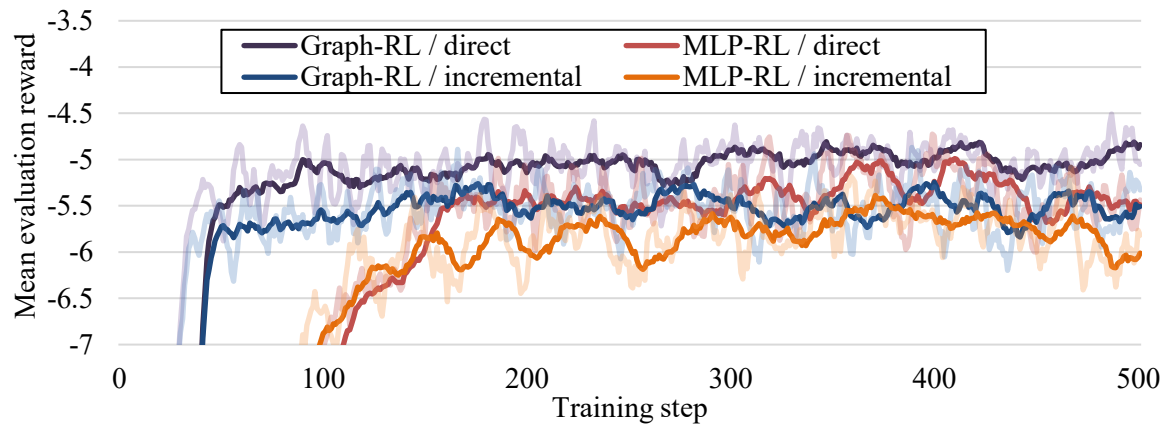


Figure 4 Training Curves under Direct and Incremental Tardiness Reward Formulations.